

W2 Python Questions

Python Questions

Exception Handling (10 Questions)

1. What is an exception in Python? How is it different from a syntax error?
 2. Explain the purpose of try-except-else-finally blocks with an example.
 3. Write code to handle a ZeroDivisionError when dividing two numbers input by a user.
 4. What is the difference between `except Exception` and `except:` in Python?
 5. How do you raise an exception manually? Provide an example scenario where you would need to raise a ValueError.
 6. Create a custom exception class called `InsufficientBalanceError` for a banking application.
 7. What is exception hierarchy in Python? Name 3 built-in exception classes and their parent classes.
 8. **Scenario:** You're reading a file and performing calculations. Write code with proper exception handling for both FileNotFoundError and ZeroDivisionError.
 9. What happens if an exception occurs in the `else` block? Will the `finally` block still execute?
 10. Explain exception chaining. How would you preserve the original exception while raising a new one?
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Object-Oriented Programming - Basics (15 Questions)

11. What is a class and an object? Explain with a real-world example.
 12. Write a simple `Car` class with attributes (brand, model, year) and a method to display car details.
 13. What is `self` in Python? Why is it needed in class methods?
 14. What is the difference between instance variables and class variables? Provide an example.
 15. Explain the `__init__` method. Is it a constructor?
 16. Create a `BankAccount` class with methods to deposit, withdraw, and check balance. Include proper validation.
 17. What is instantiation? Write code to create 3 objects of a `Student` class.
 18. Explain the difference between `instance method`, `class method`, and `static method`.
 19. Write a class with a class method that keeps count of how many objects have been created.
 20. What is encapsulation? Demonstrate with private attributes using underscore convention.
 21. How do you implement access control in Python (public, protected, private)?
 22. **Scenario:** Create a `Rectangle` class that calculates area and perimeter. Ensure width and height cannot be negative.
 23. What are getter and setter methods? Why use them instead of directly accessing attributes?
 24. Explain the `@property` decorator with an example.
 25. What is method overloading? Does Python support it natively? How can you achieve similar functionality?
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Object-Oriented Programming - Advanced (15 Questions)

26. What is inheritance? Explain with a `Vehicle` and `Car` example.
27. Write code demonstrating single inheritance where a `Dog` class inherits from an `Animal` class.
28. What is the `super()` function? When and why would you use it?
29. Explain method overriding with an example. How is it different from method overloading?
30. What is multiple inheritance? Write a class that inherits from two parent classes.
31. What is the Method Resolution Order (MRO)? How can you check it for a class?
32. **Scenario:** Create a class hierarchy: `Employee` -> `Manager` -> `Director`. Each should add specific attributes and override a `get_details()` method.
33. What is function overloading? How do you implement default arguments to achieve similar behavior?
34. Explain operator overloading. Implement `__add__` method for a `Vector` class.
35. What are abstract classes and interfaces? How do you create them using the `abc` module?
36. Write an abstract class `Shape` with an abstract method `area()`, then create concrete classes `Circle` and `Square`.
37. What are slots in Python? How do they optimize memory usage?
38. Explain the difference between `__str__` and `__repr__` magic methods.
Not covered yet --but important
39. ~~What are design patterns? Explain the Singleton pattern and implement it in Python.~~
40. **Scenario:** ~~Implement a simple Factory pattern to create different types of database connections (MySQL, PostgreSQL, MongoDB).~~

File Handling & System Operations (12 Questions)

41. What are the different modes to open a file in Python? Explain at least 5.
42. Write code to read a text file line by line and print each line.
43. What is the difference between `read()`, `readline()`, and `readlines()`?
44. How do you ensure a file is properly closed after operations? Explain two methods.
45. What is a context manager? Write code using `with` statement to write data to a file.
46. Write a program to copy contents from one file to another with proper exception handling.

47. **Scenario:** Read a CSV file containing student marks, calculate average, and write results to a new file.
 48. How do you work with binary files? Write code to read and write binary data.
 49. What is `os.path`? Write code to check if a file exists and get its absolute path.
 50. Explain the difference between `os.remove()`, `os.rename()`, and `shutil.move()`.
 51. How do you create, list, and delete directories using Python?
 52. **Scenario:** Write a script that recursively searches for all `.txt` files in a directory and counts total lines across all files.
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Standard Libraries (8 Questions)

53. What is the difference between `math` and `sys` modules? Give examples of 3 functions from each.
 54. Write code using the `random` module to generate 10 random integers between 50 and 100.
 55. How do you get command-line arguments in Python? Use the `sys` module to demonstrate.
 56. What is `datetime` module? Write code to get current date, add 30 days to it, and format it as "DD-MM-YYYY".
 57. Explain `glob` module. Write code to find all Python files in current directory.
 58. **Scenario:** Use `os` module to create a script that displays directory tree structure.
 59. What is `shutil` module used for? Write code to copy an entire directory.
 60. How do you work with file paths in a cross-platform way using `pathlib`?
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Advanced Libraries (5 Questions)

61. What is `urllib` used for? Write code to fetch content from a URL.
 62. How do you make HTTP requests in Python? Explain basic GET and POST requests.
 63. What is `unittest` module? Write a simple test case for a function that adds two numbers.
 64. Explain Test-Driven Development (TDD). What are assertions in unit testing?
 65. What is `pickle` module? Demonstrate serialization and deserialization of a Python object.
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Regular Expressions (5 Questions)

66. What are regular expressions? What is the `re` module used for?
 67. Explain the difference between `match()`, `search()`, and `findall()` in the `re` module.
 68. Write a regex pattern to validate an email address.
 69. How do you use groups in regex? Write code to extract area code from phone numbers.
 70. **Scenario:** Given a text file with mixed data, write a program using regex to extract all dates in format "DD-MM-YYYY" or "DD/MM/YYYY".
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